

Work Experience recap

Kushal Vadukul & 11/09/2026

Work experience goals



Team Structure and Workflow: See how a data science team collaborates. And runs daily projects.



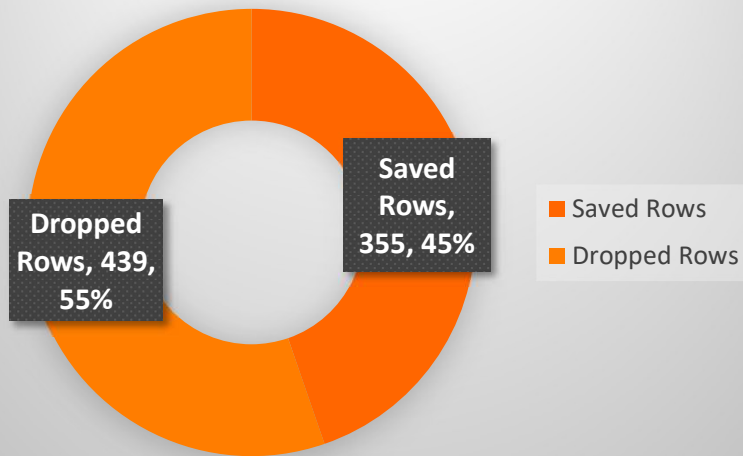
Enterprise Tools: Hands-on experience of platforms such as Databricks and working with Python.



Real World Data Science: Aimed to discover what a data scientist's job truly is and how they deal with real data trends, messy timelines and predicting future outcomes.

First task – missing values

Missing Rows



Assigned Task: Investigate the dataset and find out if there is any missing data and determine a solution for them.



Data Assessment: Identified only 794 rows with missing data out of 500,000 total rows.



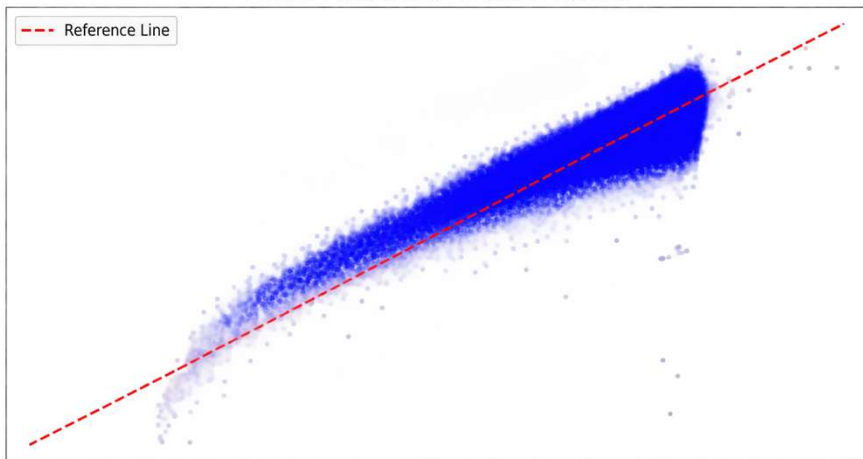
Data Filling: Using the most frequent route type to fill in the missing data, grouped by arrival_iata and applying a 75% frequency minimum.



Statistical Bias: Realising the data could be skewed, if an arrival_iata only has 1 other record, creating a false 100%(frequency).

Linear regression model

Model Predictions vs Observations



Removed tickmarks and labels from axes



Model Objective: To predict load factor from a chronological dataset ordered by departure date.



First Test: Building a Linear Regression Model using 3 raw features.



Data Leakage Problem: Creating the feature using the whole dataset not just the 80%.



Baseline Performance: Achieved a r2score of 0.83 (MSE = 36.9, MAE = 4.72)

LightGBM

Feature	R^2 Drop	MSE Increase	MAE Increase
past_loadfactor	0.81567	176.5602	6.1631
sum_standard	0.08662	18.7503	1.2188
total_flight_time	0.02964	6.4165	0.4689
sum_web_c	0.01916	4.1467	0.3224
sum_other_c	0.01287	2.7869	0.2368



Transition: Converting a linear regression model to a LightGBM (Tree Based) Model.



Label Encoding: Removing the one-hot encoded features and converting the original text column to the data type 'category'.



Creating Dominant Feature: Created past_loadfactor, which increased the r2score by 43%



Performance Change : Achieved a r2score of 0.88 (MSE = 25.4, MAE = 3.71)

Hyperparameter tuning



GridSearch Benchmark: Tested basic parameters of `n_estimators`, `max_depth` and learning rate (resulting in no score difference)



RandomizedSearch Integration: Increased to 6 parameters and successfully improved scores (r2score: 0.883, MSE: 25.2, MAE: 3.67)



Overfitting Fix: Added `min_child_samples`, `alpha` and `lambda` to stop excessively complex trees and reduce overfitting.



Bayesian Optimization: Improved by adding `colsample_bytree` and `subsample`, to make the search more flexible.



Early Stopping Results: Achieved a r2score of 0.894 (MSE = 22.9, MAE = 3.51)

Experiments

Exploring Models: Tested alternative models such as Neural Networks and XGBoost out of curiosity

Adaptability: Realising Neural Network cannot process text categories and had to reuse linear regression one hot encoding data

Experiment Conclusion: Neither model outperformed the tuned LightGBM model

Key Takeaways & What I Learnt



Forecasting: Learnt how to manage and filter data by preventing data leakage, sampling and filling in missing data and making models.



Agile Team Operations: Experienced how an agile data science team is structured and observing how projects are broken down into tickets and tracked using Jira.



Model Evaluation: Found how different models are suited to different data constraints and how a more complex model (Neural Networks) does not automatically have better results than a less complex one (LightGBM).



Cross Team Cooperation: Discovered how data scientists interact with other teams on projects and attend meetings with other teams about projects they have completed.

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Thank you!